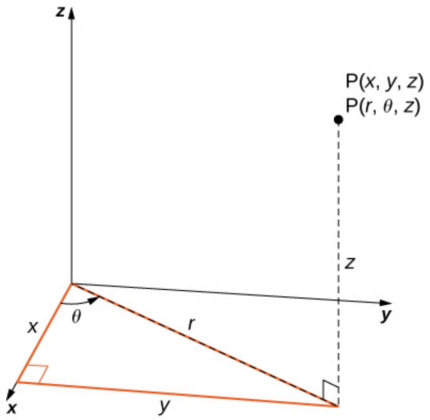
1 Cylindrical Coordinate System

In the *cylindrical coordinate system*, a point P in three-dimensional space is represented by the ordered triple (r, θ, z), where r and θ are polar coordinates of the projection of P onto the xy-plane and z is the directed distance from the xy-plane to P.



To convert from cylindrical to rectangular:

$$\boxed{x = r\cos\theta,\ y = r\sin\theta,\ z = z}$$

while to convert from rectangular to cylindrical, we use

$$\boxed{r^2 = x^2 + y^2,\ \tan\theta = \frac{y}{x},\ z = z}$$

1. (a) Plot the point with cylindrical coordinates $(2, \frac{2\pi}{3}, 1)$ and find its rectangular coordinates.
   (b) Find cylindrical coordinates of the point with rectangular coordinates $(3, -3, -7)$.

2. (a) Plot the point with cylindrical coordinates $(2, -\pi/2, 1)$ and then find its rectangular coordinates.

   (b) Find the cylindrical coordinates of the point with rectangular coordinates $(-2, 2\sqrt{3}, 3)$.

3. Describe the surface whose equation in cylindrical coordinates is $z = r$.

4. Describe the surface whose equation in cylindrical coordinates is:

   (a) $\theta = \pi/4$

   (b) $r = 5$

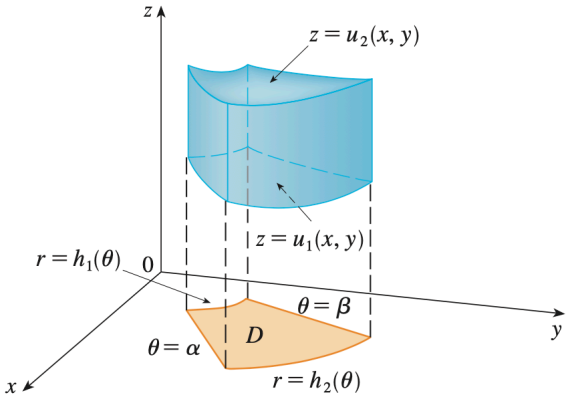
5. Sketch the solid $S = \{(r,\theta,z) : 0 \le r \le 2,\ -\pi/2 \le \theta \le \pi/2,\ 0 \le z \le 1\}$.

2 Triple Integrals in Cylindrical Coordinates

Suppose E is a Type I solid region whose projection D onto the xy-plane is conveniently described in polar coordinates. In particular, suppose that f is continuous and

$$E = \{(x,y,z) \mid (x,y) \in D, u_1(x,y) \le z \le u_2(x,y)\}$$

where D is given in polar coordinates by $D = \{(r,\theta) \mid \alpha \le \theta \le \beta, h_1(\theta) \le r \le h_2(\theta)\}$.



Then

$$\boxed{\iiint_E f(x,y,z) = \int_{\theta=\alpha}^{\theta=\beta} \int_{h_1(\theta)}^{h_2(\theta)} \int_{z=u_1(r\cos(\theta),r\sin(\theta))}^{z=u_2(r\cos(\theta),r\sin(\theta))} f(r\cos(\theta), r\sin(\theta), z)\, r\, dz\, dr\, d\theta}.$$

6. A solid E lies within the cylinder $x^2+y^2=1$, below the plane $z=4$, and above the paraboloid $z=1-x^2-y^2$. The density at any point is proportional to the distance from the axis of the cylinder. Find the mass of E.

7. Evaluate
$$\int_{-2}^{2}\int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}}\int_{\sqrt{x^2+y^2}}^{2}(x^2+y^2)dzdydx.$$

8. Evaluate
$$\int_{-3}^{3}\int_{0}^{\sqrt{9-x^2}}\int_{0}^{9-x^2-y^2}\sqrt{x^2+y^2}\,dz\,dy\,dx.$$

9. Evaluate $\iiint_E x^2\,dV$, where E is the solid that lies within the cylinder $x^2+y^2=1$, above the plane $z=0$, and below the cone $z^2=4x^2+4y^2$.